

HOW DO LANGUAGE DEVELOP?



Objectives:

- Discuss theories of language acquisition.
- Discuss prerequisites for language acquisition.
- Relate linguistic information to biology and environment.
- Summarize the nativist and behaviorist interpretation interpretation for the role of biology and behavior in nurturing language acquisition .



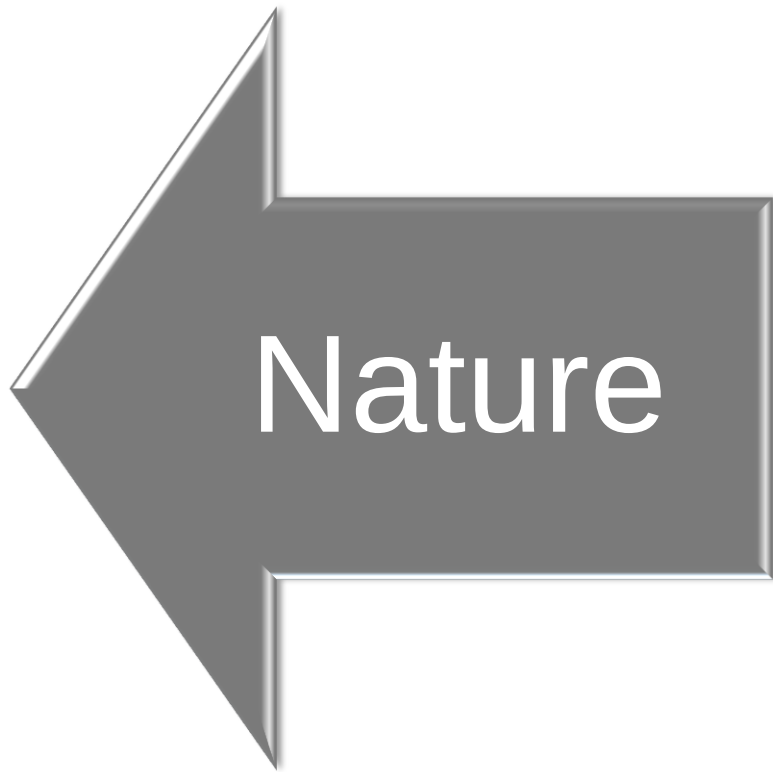
Introduction

- The capacity for acquiring language is remarkable for many reasons:
 - ✓ It is uniform throughout humanity.
 - ✓ Species specific.
 - ✓ Remarkable for its comparative speed and perfection.



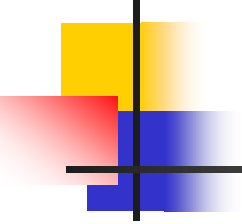
- ❖ There are many different ideas about how children learn to talk and understand language, and many controversies persist to this day.
- ❖ We are still learning about how this amazing process can occur so quickly in the first years of life.





The only language
all human beings
understand is the
language of
humanity.

Ken Bugul



Most human talents are both :



A DEFINITION OF DEVELOPMENT

Theorists differ greatly in their approach to the study of development, but all tend to agree that people develop at different rates

Development is an orderly process, and that development takes place gradually

The brain begins to lateralize soon after birth and thus to specialize certain functions in certain areas

The younger the child, the more plastic the brain the most easily the child's brain can adapt to damage.

Even though certain functions are associated with certain parts of the brain, the various parts and working systems of the brain work together to learn and to perform complex human activities and to construct understanding.

Second Edition

First Language Acquisition

Eve V. Clark



HOW CHILDREN ACQUIRE LANGUAGE?

Theories

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graph TD; A[Theories] --> B[Learning (behaviorist)]; A --> C[Nativism]; B --> D[Mixed]; C --> D;
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**Learning
(behaviorist)**

Nativism

Mixed

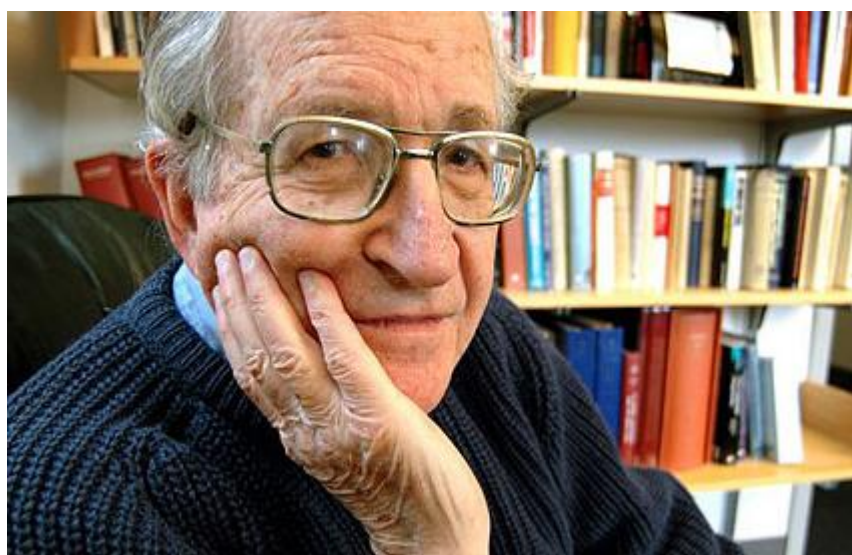


THEORIES OF LANGUAGE DEVELOPMENT

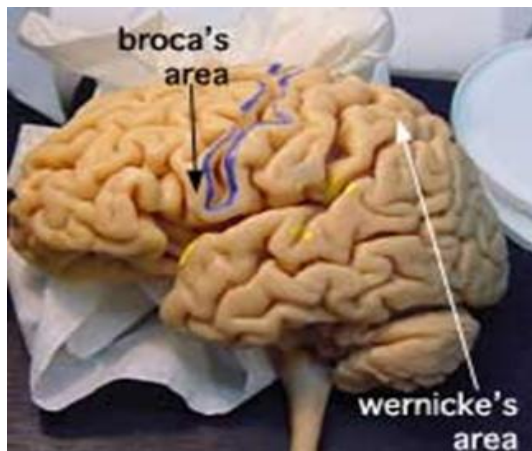
1. Learning Theory.
2. Nativist Theory.
3. Interactionist (mixed) theories.

NATIVIST THEORY

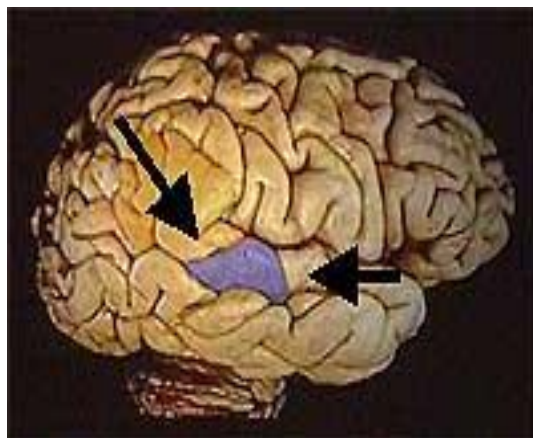
- The nativist view of language development is traced back to the work of **Noam Chomsky (1957)**.

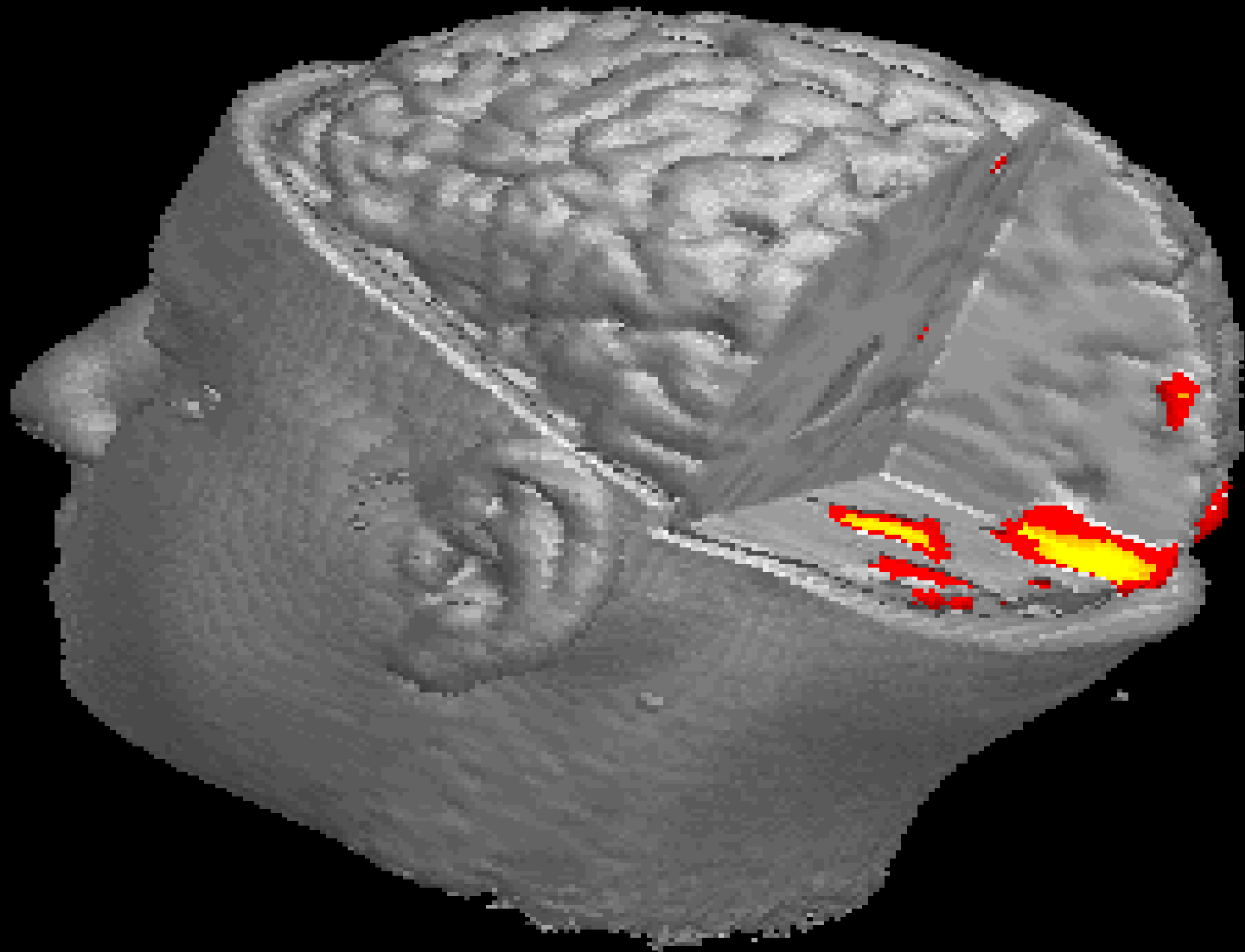






PAUL BROCA (1824-1880)
 From Schiller 1979, courtesy National
 Library of Medicine, Bethesda Md.





Plasticity of the brain

- **Plasticity:** Refers to the brain's ability to reorganise neural pathways throughout the lifespan as a result of experience.
- Put simply: The brain's ability to change with learning.
- There is a change in the internal structure of neurons, notably the synapses &
- increase in the number of synapses



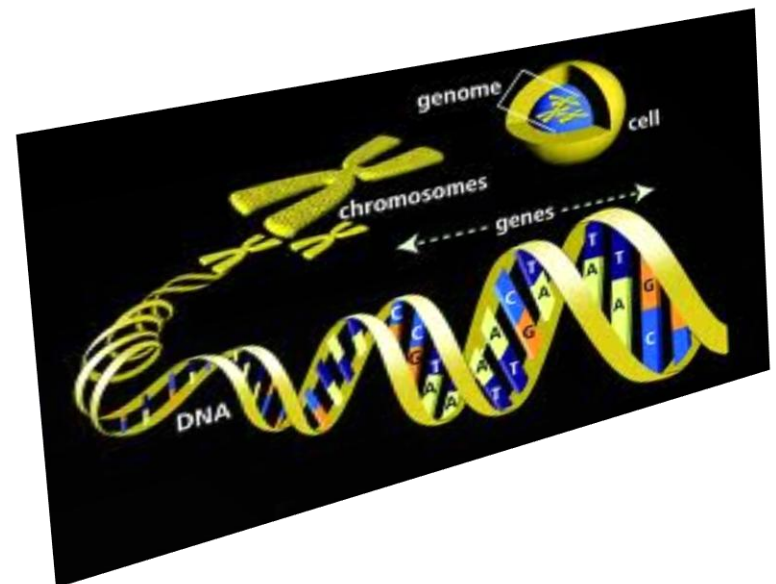
Age of acquisition effect

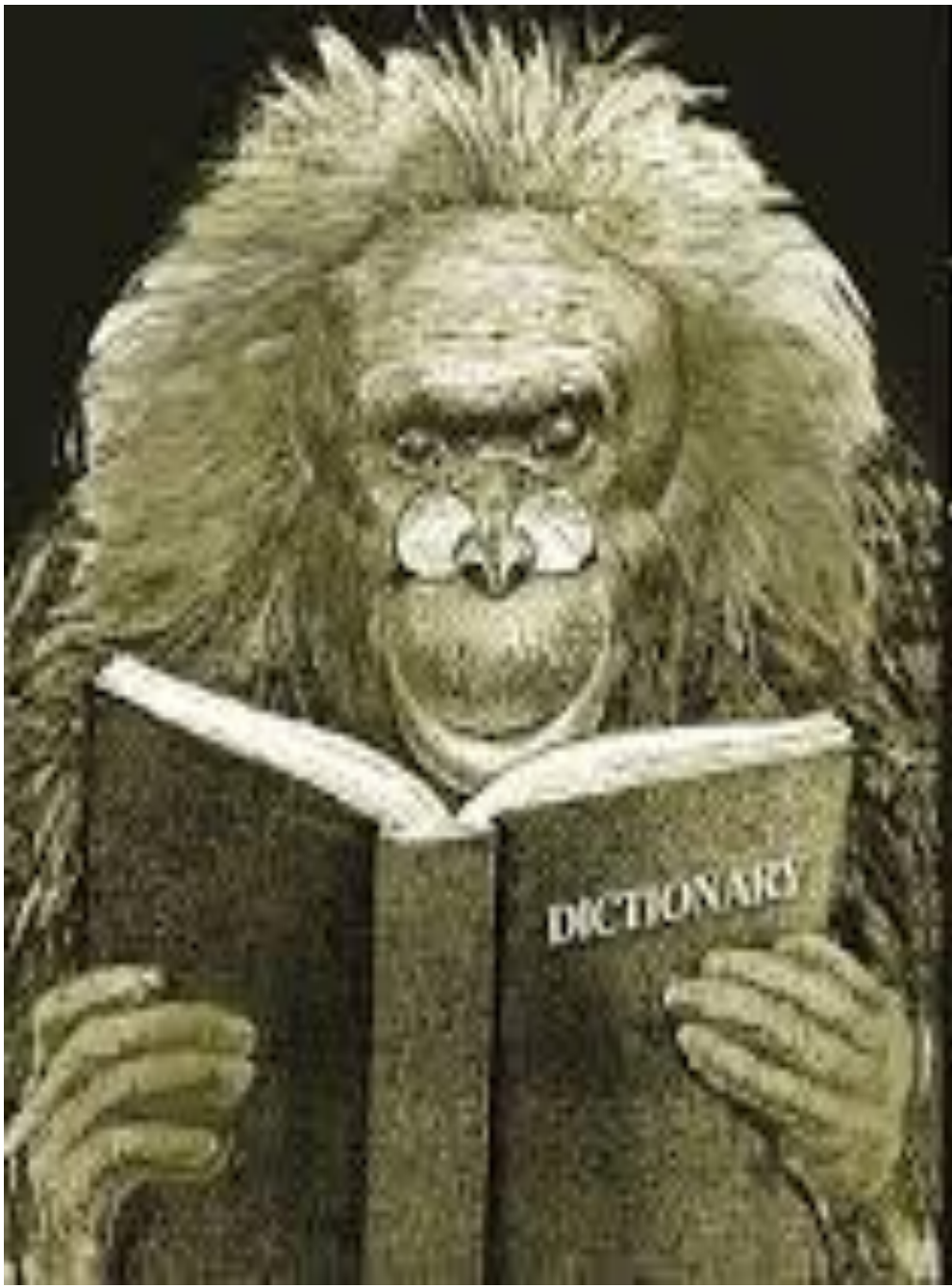


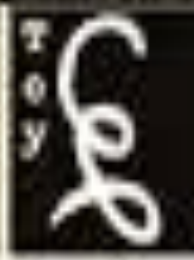


GENETIC FACTORS IN DLD:

There are currently 5 genes candidate for being associated with spoken language disorders: FOXP2, CTNAP2 (Chromosome 7), ATP2C2 and CMIP (chromosome 16), SPCH1 and KIAA0319 (CHROMOSOME 6).





 Leaf Tree	 TV Tap	 Snake	 Snow
 Greenroom	 Mary	 No	 Foot & Mouth
 Fire	 Bite	 O.J.	 Carrot
 Downstairs	 Foot	 Hot	 Toy
 Yes	 Jelly	 Dog	 Full

- So, language learning is a **stimulus –response mechanism**.
- According to this theory we start out as blank state we acquire language and its structure by experiences alone.
- These experiences is gained from our interaction with the environment.



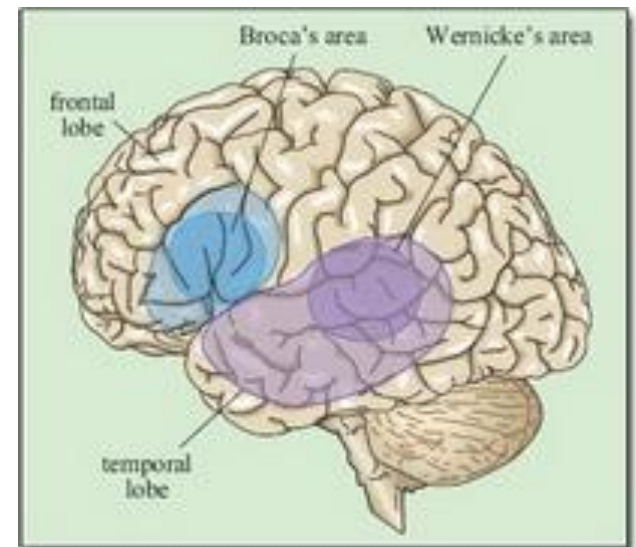
- **Chomsky** said that Because the rules which underlie a language are too complex to be acquired by children in a few short years, some aspects of language must be **innately** specified.

- In short, many aspects of language are not learned but are a part of our **biological endowment**.

All nativist theories of language development share certain elements.

- First, they assume that certain grammatical concepts are common to all languages and are therefore innate.
- Children are biologically predisposed to learn a language.

- Children come equipped with a set of innate **hypotheses** which guide their attempts to learn the rules of a language. These help to reduce the complexity of learning a language.



LEARNING THEORY.

- Accounts of language development, which emphasize that we acquire language & the structure of language using the principles of learning (through training we receive as children).
- When a child says something right he gets a reward from his parents & ultimately gets to learn it.

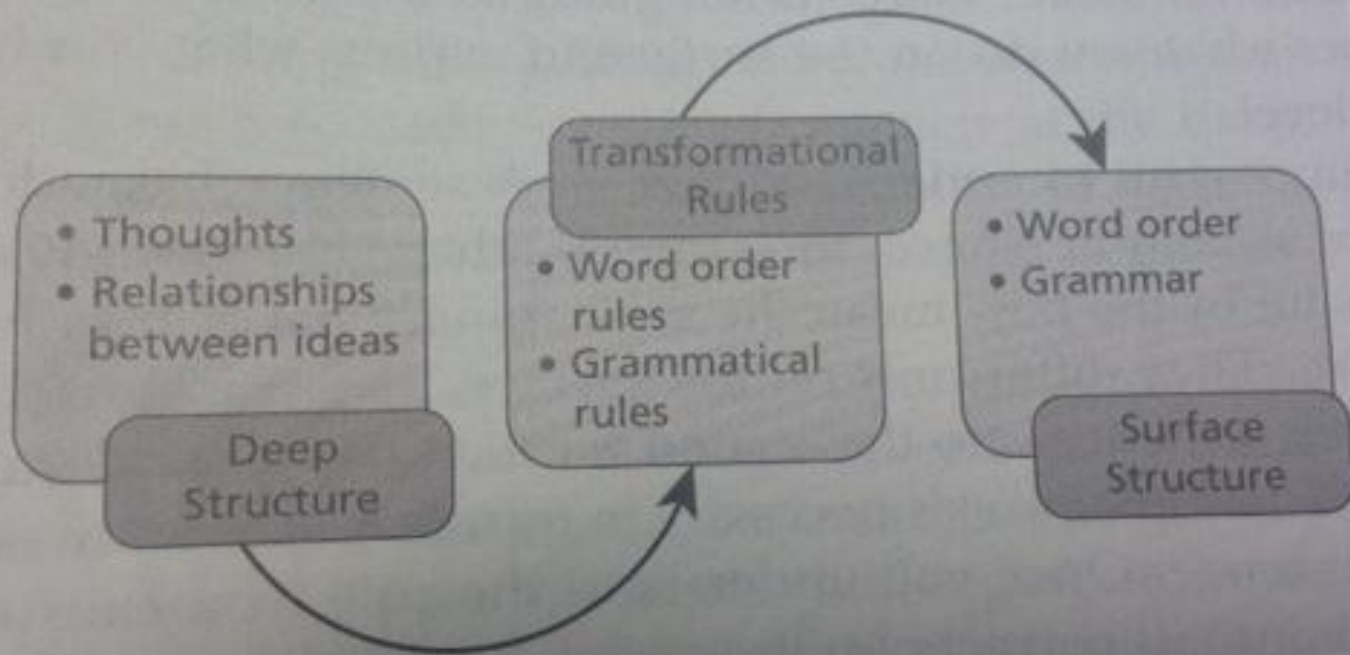
- Chomsky (1968) proposed that children come equipped with an innate mental structure -- the **language acquisition device (LAD)** -- which makes learning easier.
- According to Chomsky, the LAD contains a set of features common to all languages, which he termed a **universal grammar**.
- Universal grammar refers to the entire set of rules or linguistic parameters which specify all possible human languages.

- The learning of grammar occurs when the LAD operates on speech to abstract out the linguistic parameters which underlie the particular language used in the child's environment.



- Chomsky termed this process of determining the parameters or rules of one's native language **parameter setting**.

Figure 2.2 Transformational Generative Grammar



There are two noun phrases in our

MIXED THEORY

- ❖ It is also called the **Interactionist theories**.
- ❖ it proposes that the child's biological readiness to learn language interacts with the child's experiences with language in the environment to bring about the child's language development.

❖ **However**, in contrast to nativists, Interactionists stress the importance of both the **social support** that parents provide the young language learner, as well as the **social contexts** in which language-learning child is instructed.



❖ **Interactionist theories** are concerned with the interplay between environmental & biological factors in the process of acquiring language.

❖ Interactionists tend to view children as having a strong biological predisposition to acquire a language.

Table 2.2 ■ Tracing the Evolution of Language Acquisition Theories

Theory	Primary Architect	Essential Idea	Primary Focus
Behaviorist	Skinner (1957)	Language is learned by selective reinforcement	General
Nativist	Chomsky (1957)	Language is processed by universal and innate rules governing deep and surface structure	Syntax
Interactionist/ case grammar	Filmore (1968)	A level beneath deep structure includes universal semantic concepts that determine relationships between nouns and verbs	Semantics
Interactionist/ cognitive	Piaget (1963)	Language and thought develop as parallel processes	Thought
Interactionist/ information processing	Bates and MacWhinney (1987)	Function, not abstract grammar generates language structure	Structure by function
Interactionist/ social	Bruner (1983); Vygotsky (1962)	Language develops through social interaction and construction	Structure by use
Interactionist/ speech act	Searle (1969)	Language develops through its use in interactions	Pragmatics

CONCLUSIONS

- Language development is the result of genetics and the environment.
- The human body is built in a way that gives the creative mind many ways to communicate.
- But without external interactions, humans can not develop any language skills above what animals do

Communication Development in Infancy



INTRODUCTION

- When you ask parents about their child early language development, they will often mention the child first word or sentence.
- Although, before the child can talk. He is building a set of skills that they will need for later communication.
- **This is what is called the prelinguistic abilities.**

❖ PRELINGUISTIC SKILLS

- These skills serve as the base for later stages in the acquisition of language.

- **It can be divided into 2 domains:**

- 1- infants' perceptual ability to recognize pattern in the language input they hear around them.

- 2- They early attempt to communicate preverbally (through non word vocalizations & other gestures).

- We can say that the infant brain & sensory systems are prepared for the task of acquiring a language.
- But the child begins to acquire his language even before he is born, by the end of 3rd trimester the infant begins to hear sounds produced by his mother.



- So baby come to the world having several months of exposure to his native language.
- Babies are equipped with the ability to hear and discriminate speech sounds upon birth.
- Even they prefer to hear music or stories that they were heard in utero.

TYPICAL ORDER OF PRELINGUISTIC SKILLS IN THE 1ST YEAR

1- From 0 – 1 month:

- ✓ Turn head to look in the direction of sound.
- ✓ Prefer listening to native language.
- ✓ Prefers mother's voice than stranger.
- ✓ Discrimination of many sounds used in speech.



2- From 1- 4 months:

- ✓ Smiles when spoken to.

3- From 4-6 months :

- ✓ Show recognition of own name.

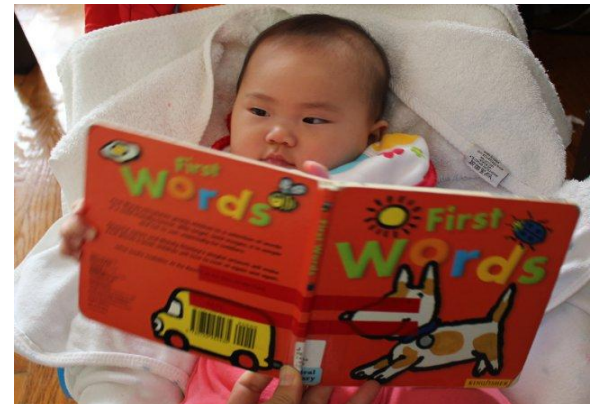
4- From 6 – 8 months:

- ✓ Able to segment a word from a fluent speech.



5- From 8 – 10 months:

- ✓ Begins to lose the ability to discriminate non- native speech sounds.
- ✓ Starts responding to own name and familiar routines (e.g. waves to “ bye-bye”).
- ✓ Show clear recognition of first word.



2- EARLY COMMUNICATION ATTEMPTS

- Communication involves more than just production & perception of sounds it involve interaction between 2 individuals
- The infants are not simply a passive recipient but they are active interactional partners & their actions affect behavior of their caregivers.

- Mothers or infant-directed speech has the following characteristics which can help infants process the speech stream:

1. a higher-pitched voice
2. longer pauses and shorter phrases;
3. A slower tempo resulting in prolonged vowel sounds.



QUESTIONS



Thank you

